



USN

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Third Semester B.E. (CS / IS / AM / CC) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21CS301 / 21IS301/ 21AM301 / 21CC301 – STATISTICS & PROBABILITY THEORY

Duration: 3 Hours

Max. Marks: 100

Note: 1. Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

2. Use of **Statistical Table** is permitted.

Unit – I

- | | Marks | BT* | CO* | PO* | | | | | | | | | | | | | | | | |
|--|-------|------|------|------|------|-------|-------|---|--------|-----|------|------|------|------|-------|-------|---|----|---|---|
| 1. a) State and prove Bayes' theorem. | 6 | L*1 | 1 | 1 | | | | | | | | | | | | | | | | |
| b) Suppose that X has distribution $N(2, 0.16)$. Evaluate $P(X \geq 2.3)$ and $P(1.8 \leq X \leq 2.1)$ | 7 | L2 | 2 | 1 | | | | | | | | | | | | | | | | |
| c) The pdf of a variate X is given by the following table:
<table border="1" style="margin: 5px auto; border-collapse: collapse;"> <tr> <td>x</td> <td>0</td> <td>1</td> <td>2</td> <td>3</td> <td>4</td> <td>5</td> <td>6</td> </tr> <tr> <td>$p(x)$</td> <td>k</td> <td>$3k$</td> <td>$5k$</td> <td>$7k$</td> <td>$9k$</td> <td>$11k$</td> <td>$13k$</td> </tr> </table> For what value of k , does this represent a valid probability distribution?
Also find i) $P(X > 4)$ ii) $P(X \geq 2)$ iii) $P(3 < X \leq 6)$ | x | 0 | 1 | 2 | 3 | 4 | 5 | 6 | $p(x)$ | k | $3k$ | $5k$ | $7k$ | $9k$ | $11k$ | $13k$ | 7 | L1 | 1 | 2 |
| x | 0 | 1 | 2 | 3 | 4 | 5 | 6 | | | | | | | | | | | | | |
| $p(x)$ | k | $3k$ | $5k$ | $7k$ | $9k$ | $11k$ | $13k$ | | | | | | | | | | | | | |
| 2. a) In a certain college 25% of boys and 10% of girls are studying mathematics. The girls constitute 60% of the students. If a student is selected at random and is found to be studying mathematics, find the probability that the student is a girl? a boy? | 6 | L1 | 2 | 1 | | | | | | | | | | | | | | | | |
| b) Suppose that a random variable X takes values 1, 2, 3, with $P(X = j) = \frac{1}{a^j} \forall j = 1, 2, 3, \dots$. Find the value of a . Also Compute i) $P(X \text{ is even})$ ii) $P(X \geq 5)$. | 7 | L2 | 1 | 1 | | | | | | | | | | | | | | | | |
| c) Find the moment generating function of a random variable X that is binomially distributed. Hence find its mean and variance. | 7 | L2 | 2 | 2 | | | | | | | | | | | | | | | | |
| 3. a) X and Y are independent. random variables X takes values 2, 5 and 7 with probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{4}$ respectively. Y takes values 3, 4 and 5 with probabilities $\frac{1}{3}, \frac{1}{3}, \frac{1}{3}$. Determine the joint p.d.f of X and Y . Also find $Cov(X, Y)$. | 6 | L1 | 1 | 1 | | | | | | | | | | | | | | | | |
| b) Find the value of c such that $f(x) = \begin{cases} \frac{x}{6} + c, & 0 \leq x \leq 3 \\ 0, & \text{elsewhere} \end{cases}$ is a pdf. Also find (i) $P(1 \leq x \leq 2)$ (ii) $E(X)$ | 7 | L2 | 1 | 2 | | | | | | | | | | | | | | | | |
| c) The length of telephone conversation in a booth has been an exponential distribution and found on an average to be 5 minutes. Find the probability that a random call made from this booth (i) ends less than 5 minutes, (ii) between 5 and 10 minutes. | 7 | L1 | 2 | 2 | | | | | | | | | | | | | | | | |

Unit – II

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|--|-----|----|---|---|---|---|-----|----|----|---|---|---|---|----|---|---|
| 4. a) Fit a straight line $y = a + bx$ to the following data by the method of least squares: | | | | | | | | | | | | | | | | |
| <table border="1" style="margin: 5px auto; border-collapse: collapse;"> <tr> <td>x</td> <td>1</td> <td>2</td> <td>3</td> <td>4</td> <td>5</td> </tr> <tr> <td>y</td> <td>14</td> <td>13</td> <td>9</td> <td>5</td> <td>2</td> </tr> </table> | x | 1 | 2 | 3 | 4 | 5 | y | 14 | 13 | 9 | 5 | 2 | 6 | L2 | 3 | 2 |
| x | 1 | 2 | 3 | 4 | 5 | | | | | | | | | | | |
| y | 14 | 13 | 9 | 5 | 2 | | | | | | | | | | | |
| b) If $\bar{X}$ is the mean of a random sample of size n from a normal distribution with mean μ and variance 100, find n such that $P(\mu - 5 < \bar{X} < \mu + 5) = 0.954$. | 7 | L2 | 4 | 2 | | | | | | | | | | | | |

- c) Find the probability that a random sample of 25 observations from a normal population with variance $\sigma^2 = 6$, will have a sample variance S^2 , (i) greater than 9.45 (ii) between 2.97 and 8.73.
5. a) Obtain the rank correlation coefficient for the following data:
- | | | | | | | | | |
|-----|----|----|----|----|----|----|----|----|
| x | 20 | 24 | 12 | 20 | 10 | 12 | 24 | 20 |
| y | 18 | 19 | 16 | 22 | 14 | 16 | 19 | 12 |
- b) Fit a curve of the form $y = a + bx + cx^2$ by the method of least squares for the following data:
- | | | | | | | | |
|-----|-----|-----|-----|------|------|------|------|
| x | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| y | 2.3 | 5.2 | 9.7 | 16.5 | 29.4 | 35.5 | 59.4 |
- c) Let X be a random variable with expectation $E(X) = \mu$ and variance $V(X) = \sigma^2$. Let $\bar{X}$ be the sample mean of a random sample of size n . Then prove that :
- (a) $E(\bar{X}) = \mu$.
 (b) $V(\bar{X}) = \sigma^2/n$.
6. a) Show that sample mean is an unbiased estimator whereas sample variance is not. Show that if $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$, then it is unbiased.
- b) Find the correlation coefficient between x and y for the given values. Find also the two regression lines.
- | | | | | | | | | | |
|-----|---|---|----|----|----|----|----|----|----|
| x | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| y | 9 | 8 | 10 | 12 | 11 | 13 | 14 | 16 | 15 |
- c) Suppose that a random sample of size 100 is taken from the inner diameters of certain lengths of pipes. If the mean and standard deviation of such measurements are 34.1 and 1.5 inches respectively, find the probability that the sample mean will be between 34 and 34.3.

Unit – III

7. a) Define probability vector. Find the unique fixed probability vector for the regular stochastic matrix $\begin{bmatrix} 0 & 2/3 & 1/3 \\ 1/2 & 0 & 1/2 \\ 1/2 & 1/2 & 0 \end{bmatrix}$.
- b) Let $\{X_n\}$ be the stochastic process with state space $S = \{1, 2, 3\}$. Given transition probability matrix of a Markov Chain $P = \begin{bmatrix} 0.1 & 0.5 & 0.4 \\ 0.6 & 0.2 & 0.2 \\ 0.3 & 0.4 & 0.3 \end{bmatrix}$ with initial probability distribution $p^0 = (0.7, 0.2, 0.1)$. Find i) $P(X_2 = 3)$ ii) $P(X_1 = 3)$ iii) $P(X_3 = 2, X_2 = 3, X_1 = 3, X_0 = 2)$.
- c) In a cascade of binary communication channels, the symbol 1 and 0 are transmitted in successive stage. In any stage, the probability of a transmitted 1 is received as 1 is 0.75 and the probability that a 0 is received as 0 is 0.5. Find the probability that:
- i) 1 transmitted in the first stage is received correctly and 0 transmitted in the first stage is received as 1 after third stages.
 ii) If the probability of transmitting a 1 in the initial stage is $5/8$, find the probability that 1 is received and a 0 received after three stages.
8. a) A salesman's territory consists of 3 cities A, B and C. He never sells in the same city on successive days. If he sells in the city A, then the next day he sells in city B. However if he sells in either B or C, then the next day he is twice as likely to sell it in city A as in other city. Show that in the long run he sells 40% of the time in city A, 45% of the time in city B and 15% of the time in the city C.

- b) Let $\{X_n\}$ be the stochastic process with state space $S=\{1,2,3,4\}$ and uniform $P(0)$. The transition probability matrix is

$$P = \begin{bmatrix} \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{2} & 0 & \frac{1}{4} & \frac{1}{4} \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix}$$

- a) Draw the state transition diagram
 b) Find $P[X_5 = 3, X_3 = 2 | X_1 = 3]$
 c) Find $P[X_3 = 2, X_2 = 1 | X_1 = 1]$
 c) At a service center customers arrive at the rate of 10 per hour and are served at the rate of 15 per hour. Their arrival follows Poisson distribution and service is exponentially distributed. Find the average length and average waiting time in the system.

7 L2 5

7 L1 5

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
